Internship Management System

Student Lifecycle - Step-by-Step Guide (how a student is created, applies, gets approved, and all data interactions)

Introduction

The platform connects Colleges, Faculty/Mentors, Companies, Supervisors and Students. A student enters the system through the College, then moves through application, approval, mentor assignment, supervisor assignment, attendance, task and progress tracking, and finally certification. This document explains every step, who does what, and how the data is calculated. Demo credentials: all portal accounts use password Demo@12345 (admin superuser: admin / Admin@12345).

Step 1 - College creates the student account

- Log in to the College portal (college_admin / Demo@12345).
- Go to Students and click Add Student.
- Enter student details (name, email, department, year) and submit.
- The system creates the login account with the default password Demo@12345 (or a generated one).
- Result: the student appears in the College dashboard list. It does NOT appear on the Faculty, Company or other dashboards yet.

Step 2 - Student logs in and completes the profile

- Student opens /student/login/ and logs in (username e.g. student1, password Demo@12345).
- The student dashboard shows their profile, allocation status and available internship listings.
- The student can upload documents (offer letter, joining report, etc.).

Step 3 - Student applies to an internship

- From the student dashboard the student browses active internships and clicks Apply.
- An internship application is created with status Pending.
- The student can have one active allocation at a time.

Step 4 - College reviews and approves the application (allocation)

- The College dashboard shows pending applications.
- The college reviews and approves the application.
- Approval creates an ACTIVE allocation linking Student -> Internship (-> Company). Start/end dates are set.
- Notifications inform the student (and company) that the allocation is live.

Step 5 - Student selects a Faculty Mentor (new feature)

- On the student's Mentor page, the system lists all Faculty members of the student's college.
- The student selects a mentor; the assignment is stored in the DB (FacultyAssignment, active=True).
- Changing the mentor deactivates the old assignment and creates a new one - no duplicate active mentors (unique constraint).

- Result: the student appears on the selected mentor's dashboard (Faculty -> My Students).
- A student from another college cannot pick that college's faculty (403 PermissionDenied).

Step 6 - Company verifies a Supervisor; Supervisor marks attendance (new feature)

- Each internship has CompanySupervisors (contact records) with a hashed PIN/password. After allocation, the company links an intern to a supervisor via Assign Supervisor - stored as SupervisorAssignment (active=True, one active supervisor per intern/internship).
- Supervisors get a login account (username supervisor1..5, password Demo@12345, PIN 1234) and open /supervisor/login/.
- Company Dashboard -> Intern Attendance -> Select Supervisor -> Enter PIN -> verify -> Show All Assigned Interns. The company is now scoped to that supervisor (session-based, must belong to the same company).
- The supervisor view lives at /company/supervisor//... and only lists interns assigned to that supervisor; Attendance -> Mark Attendance is scoped the same way.
- Each Attendance record is stored with the supervisor and marked_by fields. Duplicates for the same student/date are rejected.
- The company exits the supervisor view via an Exit button that clears the session (no direct attendance management outside the PIN-verified flow).

Step 7 - Company assigns tasks via the supervisor; supervisor reviews; faculty monitors (new feature)

- Company portal -> Tasks -> Assign Task. The form requires BOTH the intern (student) AND a supervisor (mandatory).
- The task is stored with assigned_by = company user and supervisor = selected supervisor; the supervisor is also auto-linked to the intern via SupervisorAssignment.
- A task without a supervisor is rejected by validation.
- The supervisor view -> Tasks lists tasks of their interns and can Approve / Reject SUBMITTED tasks.
- Faculty dashboard shows ALL tasks (title, description, student, supervisor, status, progress, start, due) VIEW-ONLY - no assign, edit, review or reassign buttons.

Step 8 - Faculty reviews work logs and weekly progress

- Students submit Daily Work Logs and Weekly Progress reports.
- Faculty reviews each submission (approve / revise) - this is the faculty's review responsibility.
- Reviews and auto-calculations update the progress summary on the faculty and college dashboards.

Step 9 - Data calculations happen automatically

- Attendance percentage = present / working days (excludes Sundays), shown per student.
- Weekly progress average = mean of the student's weekly progress reports.
- Pending/overdue task counts are computed from task status vs due date.
- If attendance drops below the warning threshold, the college gets attendance warnings.

Step 10 - Documents, completion and certificate

- Students upload required documents (offer letter, joining and completion reports) in their portal.

- College or company uploads/generates the completion certificate with a certificate number (e.g. INT-2026-000002).
- Anyone can verify a certificate publicly at /certificates/verify// (shows student and company names).

Step 11 - Interactions between dashboards (summary)

- Adding one student: it appears ONLY on the college dashboard by default.
- After Step 5 (mentor selection) it also appears on the chosen faculty dashboard.
- After Step 4 (application approval) + Step 6 (supervisor assignment) it appears on the company dashboard and in the supervisor view (Assigned Interns).
- Workflow: Company -> Supervisor -> Intern -> Task. Faculty and college only view task progress.
- A company needs to re-verify a supervisor PIN to enter the supervisor view; Supervisor A never sees Supervisor B's interns.

Role-Based Access Control

• **College** - Creates student accounts, approves applications, sees analytics and reports, uploads certificates.

• **Faculty / Mentor** - Sees assigned students, reviews work logs and weekly progress, and views task progress for all of their students (view-only - cannot assign or review tasks).

• **Company** - Posts internships, manages students and supervisors, assigns tasks (with mandatory intern + supervisor), requests completion, uploads certificates. Enters attendance via Intern Attendance -> PIN verify -> supervisor view.

• **Supervisor** - Supervisor view at /company/supervisor//...: marks attendance for their own assigned interns, analytics/reports with CSV/XLSX/PDF export, approves/rejects their task submissions.

• **Student** - Applies to internships, selects a mentor, works on tasks, logs progress, uploads documents, views own attendance/certificate.

Verification Summary (Deliverable 6)

Check	Result
manage.py check	System check identified no issues (0 silenced)
manage.py test	Ran 114 tests ... OK (all 14 apps)
Smoke check	All portal pages return 200 incl. supervisor view pages under /company/supervisor//; company PIN verify flow works (correct PIN -> students list, exit -> 403 on next access, wrong PIN rejected); removed /supervisor/dashboard/ returns 404; faculty /faculty/tasks/ view-only
DB migrations	5 new migrations: accounts.0002 (SUPERVISOR role), internships.0002 (CompanySupervisor.user + SupervisorAssignment), internships.0003 (CompanySupervisor.pin), tasks.0002 (Task.supervisor), attendance.0002 (marked_by/supervisor)

Files / Components Changed

- accounts/models.py - Role.SUPERVISOR; get_profile/display_name handle supervisor_profile
- accounts/decorators.py - supervisor_required + SupervisorRequiredMixin; get_role_dashboard_url() sends supervisors to supervisors:dashboard
- accounts/forms.py - SupervisorAuthenticationForm; accounts/views.py - SupervisorLoginView; accounts/urls.py - /supervisor/login/
- accounts/factories.py - create_supervisor(); seed_demo.py - supervisor accounts, assignments, marked attendance (--force flag)
- internships/models.py - CompanySupervisor.user (OneToOne); SupervisorAssignment model
- internships/services.py - assign_supervisor() service
- tasks/models.py - Task.supervisor FK; tasks/services.py - create_task accepts supervisor
- attendance/ - Attendance.supervisor + marked_by; attendance/services.py - mark_attendance accepts marked_by/supervisor; attendance/reports.py - 10 report types; attendance/analytics.py - intern/supervisor analytics
- supervisors/ - access.py (PIN session + supervisor_template_required decorator), context_processors.py, views (dashboard, students, attendance, analytics, reports + CSV/XLSX/PDF export, tasks, progress, notifications), urls (app_name, /company/supervisor//), tests
- companies/ - attendance_select + attendance_exit (PIN verify/exit), SupervisorVerifyForm, /company/attendance/ URLs; task_assign requires intern + supervisor; student_assign_supervisor view
- internwebsite/urls.py - company/supervisor/ mounted before company/
- faculty/ - removed task_assign/task_review views, urls, forms, template; tasks.html now view-only
- templates/supervisors/ - dashboard, students, attendance, analytics, reports, tasks, progress, notifications; partials/sidebar.html - Intern Attendance + Supervisor View nav blocks
- accounts/home.html - Supervisor portal login card
- README.md - five portals + PIN-verified Intern Attendance flow documented